# EDNS 491 & 492: Capstone Design I & II

## Project Advisor's Guide

Academic Year 2025–2026

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**Purpose of this guide:** This document is a practical reference for Project Advisors (PAs) supervising teams in Capstone Design. It covers the PA role, sprint-by-sprint facilitation notes, design review scripts, grading guidance, and strategies for managing team dynamics. Each section is self-contained—you can read only the sections relevant to your immediate need. For deep technical and pedagogical reference, consult the Course Text chapters cited in each section.

**Four documents work together**: (1) this PA Guide (instructor-facing reference), (2) the Course Text (22-chapter deep reference for students and PAs), (3) the Student Guide (sprint-level navigation for students), and (4) Canvas (assignments, rubrics, deadlines, and the Course Syllabus).

**A note on terminology:** This guide uses Colorado School of Mines terminology—PA (Project Advisor), CF (Capstone Faculty), CP (Client Partner), CLT (Capstone Leadership Team), InnoHub (Innovation Hub fabrication space). At other institutions, the PA role may be called "faculty mentor," "project advisor," "capstone instructor," or "section leader." The principles are the same; adapt the labels to your context.

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# 1 Purpose and Document Ecosystem

This guide exists because effective project advising requires more than technical expertise—it requires understanding of the course structure, grading philosophy, common failure modes, and when to intervene versus when to let students struggle productively. Whether you are a first-time PA or a veteran, this document provides a shared baseline for how we run Capstone Design.

## 1.1 The Four-Document Ecosystem

The Capstone Design program is supported by four coordinated documents. Each serves a distinct audience and purpose:

1. **Course Text** (22 chapters, 5 parts) — The comprehensive reference for both students and PAs. Covers engineering design methodology, project management, professional skills, safety, and compliance. When students ask "why do we have to do this?" the Course Text provides the pedagogical justification and worked examples.
2. **Student Guide** — A sprint-by-sprint navigation aid for students. Tells them what to focus on each sprint, common pitfalls to avoid, and deliverable checklists. Written in a direct, student-facing tone.
3. **Canvas** — The authoritative source for assignments, rubrics, deadlines, and the Course Syllabus. All grading policies, late-work rules, and academic integrity standards live here. The Course Syllabus is the contractual document.
4. **PA Guide** (this document) — Instructor-facing reference covering facilitation, grading calibration, team management, and design review scripts. Not intended for student distribution.

When a student asks a question, point them to the right document. Course mechanics and deadlines: Canvas. Design methodology and theory: Course Text. Sprint-level priorities: Student Guide. Resist the urge to re-explain what is already documented—teach students to use the resources.

## 1.2 Quick Start: Your First Week as a PA

If you are new to the PA role, complete the following during your first week:

1. **Read the Course Syllabus on Canvas.** Understand grading weights, attendance policies, and academic integrity expectations. You will be asked about these.
2. **Skim the Student Guide.** Understand what students are being told. This prevents contradictory messaging.
3. **Read Sections 2–4 of this guide.** These cover your role, the sprint structure, and design review facilitation—the core of your weekly work.
4. **Review your team assignments.** Note the project type (Build, Paper, Competition, or Hybrid), the client organization, and any special constraints (e.g., ITAR, competition deadlines).
5. **Attend the PA orientation meeting.** The CF and CLT Director will brief you on this semester's schedule, any policy changes, and administrative procedures.
6. **Set up your meeting cadence.** Schedule a recurring weekly or biweekly meeting with each of your teams. Consistency matters more than frequency.

7. **Introduce yourself to your teams' clients.** A brief email establishing that you are the PA and providing your contact information sets the right tone.

Before the first sprint review, review the Course Text chapters covering the current sprint's topics. You do not need to read all 22 chapters—but you should always be at least one sprint ahead of your teams in terms of content familiarity.

# 2 The PA Role

*Reference | Course Text: Chapters 1, 11, 13, 17*

At other institutions, the PA role may be called "faculty mentor," "project advisor," "capstone instructor," or "section leader." Regardless of title, the function is the same: you are the primary point of contact between the course and the team, and your judgment is the most influential factor in each student's experience.

## 2.1 Core Responsibilities

1. **Mentoring.** Guide teams through the design process without doing the design for them. Ask questions that expose gaps in reasoning. Suggest approaches without prescribing solutions.
2. **Grading and Assessment.** Provide PA input on team and individual grades. Evaluate deliverables for engineering quality, not just completeness. Complete Individual Performance Evaluations on schedule.
3. **Gate-Keeping.** Verify that teams meet minimum readiness criteria before advancing through design reviews (PDR, CDR, FDR). A team that is not ready for CDR should not present at CDR.
4. **Client Liaison.** Maintain awareness of the client relationship. Intervene if communication breaks down. Attend client meetings when invited or when the relationship needs support.
5. **Safety Oversight.** Review and approve Hazard Assessments before fabrication begins. Escalate safety concerns to the CLT Director immediately.

## 2.2 Relationship with CF, CLT Director, and Other PAs

- The **Capstone Faculty (CF)** sets course policy, owns the syllabus, and makes final grade determinations. PAs provide input; the CF decides.
- The **CLT Director** manages program logistics, client relationships at the organizational level, and facilities (InnoHub, labs). Escalate facility, safety, or client-relationship issues to the CLT Director.
- **Other PAs** are your peers. Calibrate grading expectations by comparing notes at PA meetings. If you are unsure whether a deliverable meets the bar, ask a fellow PA for a second opinion.

## 2.3 Time Commitment

Expect to invest approximately **3–5 hours per week per team**, distributed roughly as follows:

| Activity | Approx. Hours/Week |
|---|---|
| Sprint review meetings (preparation + attendance) | 1.0–1.5 |
| Deliverable review and feedback | 1.0–1.5 |
| Email, Slack, and ad-hoc student questions | 0.5–1.0 |
| PA coordination meetings and calibration | 0.25–0.5 |
| Design review preparation (PDR/CDR/FDR weeks) | +2.0–3.0 |

Time investment increases during design review sprints and decreases during steady-state execution sprints (Sprints 9–11).

### 2.4 Authority and Boundaries

**What does the PA decide vs. what does the CF decide?**

**PA decides:**
- Whether a team is ready for a design review (gate-keeping)
- Day-to-day mentoring approach and meeting cadence
- Sprint review feedback and coaching priorities
- Input on individual and team grade recommendations

**CF decides:**
- Final grades and grade disputes
- Course policy exceptions (late work, extensions, accommodations)
- Academic integrity cases
- Team reassignments or project changes
- Performance Improvement Plans (PIPs)

Do not make promises about grades. You may say "this is strong A-level work" or "this needs significant improvement to pass," but final grade determination is the CF's responsibility. Over-promising grades creates conflicts that are difficult to resolve.

Establish your communication norms early. Tell your teams how you prefer to receive questions (email, Slack, office hours), your typical response time, and what constitutes an emergency worth a same-day message. Setting expectations in Sprint 0 prevents friction later.

## 3 Sprint-by-Sprint Facilitation Notes

*Reference | Course Text: Chapters referenced per sprint below*

This section provides a half-page summary for each sprint: what to look for, red flags, key deliverables, and suggested questions for sprint reviews. Use this as your pre-meeting preparation checklist.

### 3.1 Sprint 0: Project Expo, Team Formation, Client Kickoff

*Weeks 1–2 | Course Text: Chapters 1, 2*

**Health indicators:** Team has contacted the client. Meeting logistics are confirmed. Team members have introduced themselves and begun discussing roles.

Team has not contacted the client by end of Week 1. A team that cannot send an email will struggle with every subsequent communication milestone.

**Key deliverables:** Project Selection Survey, client introduction email with team slide, initial meeting scheduled.

- What does the client actually need—not what do they say they want?
- What are the top three unknowns about this project right now?
- How will you prepare for the kickoff meeting?

### 3.2 Sprint 1: Client Discovery, Problem Framing

*Weeks 3–4 | Course Text: Chapters 2, 3*

**Health indicators:** Kickoff meeting completed. Meeting minutes distributed. Team can articulate the problem in their own words (not just echoing the client brief).

Team is already proposing solutions. Sprint 1 is for understanding the problem, not solving it. If you hear "we're going to use an Arduino," redirect to "what problem are you solving?"

**Key deliverables:** Kickoff meeting minutes, initial needs/goals documentation, stakeholder identification.

- Who are the stakeholders beyond the primary client contact?
- What constraints has the client stated explicitly? What constraints are implied?
- What does success look like from the client's perspective?

### 3.3 Sprint 2: Requirements Development, Research

*Weeks 5–6 | Course Text: Chapters 3, 4*

**Health indicators:** Requirements are written in "shall" statement form with measurable acceptance criteria. The team distinguishes between requirements and specifications. Background research is underway.

Requirements that read like specifications ("The system shall use a 12V motor") instead of functional needs ("The system shall generate a minimum torque of 5 N·m"). This is the single most common Sprint 2 error.

**Key deliverables:** Draft SDRD (System Design Requirements Document), literature review progress, concept generation begins.

- How will you verify each requirement? If you cannot test it, rewrite it.
- What engineering standards apply to this project?
- What did your background research reveal that surprised you?

## 3.4 Sprint 3: Concept Development, Client Approval

*Weeks 7–8 | Course Text: Chapters 3, 4, 17*

**Health indicators:** Multiple concepts generated (target ≥15 ideas before down-selection). Decision matrix uses weighted criteria with sensitivity analysis. Client has reviewed and approved the selected approach.

Only one concept considered, or the decision matrix is clearly rigged to justify a predetermined choice. Ask the team to vary the weights by ±20% and report whether the winner changes.

**Key deliverables:** Decision matrix with sensitivity analysis, client approval of approach, engineering standards identification.

- What were the top three concepts you eliminated, and why?
- If you change the top-weighted criterion, does your winner change?
- What is the client's biggest concern about the selected concept?

## 3.5 Sprint 4: Preliminary Design, PoC Planning

*Weeks 9–10 | Course Text: Chapters 4, 6, 7*

**Health indicators:** SOW is drafted with clear scope boundaries (including an "Out of Scope" section). The Proof of Concept (PoC) targets the project's key technical risk, not the easiest thing to build.

PoC plan that tests something the team already knows works. The purpose of a PoC is to retire the *highest* technical risk. If the team's PoC does not make them nervous, it is probably targeting the wrong risk.

**Key deliverables:** Draft SOW, PoC plan, initial project schedule (Gantt or backlog).

- What is the single biggest technical risk in this project?
- Does your PoC directly address that risk?
- What is explicitly out of scope, and has the client agreed?

## 3.6 Sprint 5: PDR Preparation, PoC Execution

*Weeks 11–12 | Course Text: Chapters 7, 8*

**Health indicators:** PoC is built and tested (not just planned). PDR report draft is under internal review. The team has scheduled a mock PDR.

Team skips the mock PDR. Mock PDRs consistently improve final PDR quality. If a team claims they "don't have time," they especially need one.

**Key deliverables:** PoC results, PDR report draft, PDR presentation draft, mock PDR completed.

- What did the PoC teach you? Did anything fail, and what did you learn from the failure?
- Walk me through your PDR presentation flow in 60 seconds.
- What question are you most afraid a reviewer will ask at PDR?

## 3.7 Sprint 6: PDR, Long-Lead Procurement, Semester Wrap

*Weeks 13–14 | Course Text: Chapters 8, 14, 15*

**Health indicators:** PDR is complete. Feedback has been documented and an action plan exists. Long-lead items are ordered or have purchase orders submitted. Budget is realistic.

No procurement activity after PDR. If the team has not ordered long-lead components (PCBs, custom parts, specialty materials) by end of Sprint 6, those items will not arrive in time for SDII fabrication.

**Key deliverables:** PDR presentation and report, PDR feedback action plan, procurement initiation, semester reflection.

- What were the top three weaknesses identified at PDR, and what is your plan to address each?
- What is your longest lead-time item, and when will it arrive?
- What will you accomplish over the break to stay on track?

Complete Peer Evaluation #2 and Individual Performance Evaluation #1 before the semester ends. Submit your PA grade input to the CF by the posted deadline.

### 3.8 Sprint 7: SDII Kickoff, Detailed Design, Hazard Assessment

*Weeks 15–16 | Course Text: Chapters 9, 12, 21*

**Health indicators:** Team has addressed PDR feedback. Detailed design is progressing (CAD, schematics, analysis). Hazard Assessment is drafted. Fabrication plan is realistic.

Team wants to start fabrication without a completed Hazard Assessment. This is a hard stop. No fabrication may begin until the Hazard Assessment is reviewed and approved by the PA.

**Key deliverables:** PDR feedback closure, detailed design progress, Hazard Assessment, updated project schedule.

- Which PDR feedback items have you closed? Which are still open?
- Walk me through your Hazard Assessment. What is the highest-severity hazard?
- Is your fabrication timeline realistic given InnoHub availability and material lead times?

### 3.9 Sprint 8: Fabrication Begins, CDR Preparation

*Weeks 17–18 | Course Text: Chapters 9, 12, 14*

**Health indicators:** Fabrication is underway (not just planned). BOM is complete—including fasteners, consumables, and wiring. CDR report draft is in progress.

BOM missing fasteners and consumables. Have the team walk through the CAD assembly and identify every component that needs a BOM line. If it is visible in the model, it needs a part number and a cost.

**Key deliverables:** Fabrication progress, complete BOM, CDR report draft, CDR presentation draft.

- Show me your BOM. Is every component accounted for, including fasteners?
- What is your integration plan—how will subsystems come together?
- What is your budget status? Any overruns or pending purchases?

### 3.10 Sprints 9–11: Build, Integration, Testing

*Weeks 19–24 | Course Text: Chapters 6, 9, 12*

**Health indicators:** Subsystem testing is occurring in parallel with fabrication. Integration is incremental (one subsystem at a time), not big-bang. V&V testing has started—not deferred to Sprint 12.

All testing deferred to "after we finish building." Testing should begin as soon as the first subsystem is functional. Every week of testing margin is one redesign cycle. Teams that leave all testing to Sprint 12 run out of time to fix problems.

**Key deliverables:** Subsystem test results, integration progress, V&V test execution, updated VCRM.

- Which subsystems have been tested independently? What were the results?
- What is your integration sequence, and what are the interface risks?
- How many requirements have you verified so far? How many remain?

During these sprints, shift your focus from design guidance to project management oversight. Ask to see the Gantt chart or backlog at every meeting. Teams that lose schedule visibility in Sprints 9–11 are the ones that scramble at FDR.

### 3.11 Sprints 12–13: Final Testing, Documentation, FDR Prep

*Weeks 25–28 | Course Text: Chapters 6, 10, 18*

**Health indicators:** V&V testing is substantially complete. VCRM shows all requirements dispositioned. FDR report sections are being written concurrently with testing (not deferred to the last week). Documentation debt is minimal.

Documentation debt is the number one indicator of a project in trouble at this stage. If the team has been building for six sprints without writing report sections, they will produce a rushed, low-quality FDR report. Check documentation status at every meeting.

**Key deliverables:** Completed V&V results, VCRM with all requirements dispositioned, FDR report draft, FDR presentation draft.

- Show me the VCRM. Which requirements are verified, failed, waived, or deferred?
- For any failed requirements, what is the disposition and path forward?
- How many pages of the FDR report are drafted? How many remain?

### 3.12 Sprint 14: FDR, Design Showcase, Course Closeout

*Weeks 29–30 | Course Text: Chapters 10, 19*

**Health indicators:** FDR report is complete and sent to the client. Presentation is rehearsed. Demo hardware is tested and working. The team can articulate what they would do differently.

Incomplete VCRM at FDR. Every requirement must be dispositioned: Verified, Failed (with justification and recommendation), Waived (with client agreement), or Deferred (with rationale). An incomplete VCRM signals that the team did not complete the design cycle.

**Key deliverables:** FDR presentation and report, Design Showcase poster and demo, final client handoff, course evaluations.

- If you had one more sprint, what would you change?
- What was the most significant technical challenge, and how did you resolve it?
- What advice would you give to next year's teams?

Complete all remaining Peer Evaluations and Individual Performance Evaluations. Submit final PA grade input to the CF by the posted deadline. Ensure client handoff documentation is complete.

# 4 Facilitating Design Reviews (PDR/CDR/FDR)

*Reference | Course Text: Chapters 8, 9, 10*

Design reviews are the primary gate-keeping mechanism in Capstone Design. Your role as PA is to ensure that reviews are rigorous, fair, and educational. A well-run design review teaches students more about engineering judgment than any lecture.

## 4.1 General Facilitation Principles

- Start with a brief (2-minute) context reset: project name, client, problem statement. Reviewers who are not the PA need this.
- Let the team present without interruption for the first 5–8 minutes, then open Q&A by section.
- Direct questions to specific team members, not just the presenter. This surfaces individual understanding and draws out quieter members.
- End every review with two categories of feedback: (1) strengths to preserve and (2) specific items to address before the next gate.
- Document all feedback in writing. The team's scribe should capture it, but verify the list before adjournment.

## 4.2 Mock PDR Reviewer Script

Use the following questions as a starter set for mock PDRs. Organize your questioning by topic area, and adapt based on the specific project.

**Problem Definition and Requirements**

1. What is the core problem you are solving, stated in one sentence?
2. How did you validate that this is the right problem? Did the client confirm?
3. Walk me through your top five requirements. How will you verify each one?
4. Which requirement is the hardest to meet, and why?
5. Are there requirements that conflict with each other? How will you resolve the trade-off?

**Concept Selection and Design**

6. How many concepts did you generate before down-selection?
7. What were the top two alternatives to your chosen concept? Why did you reject them?
8. Vary your decision matrix weights by 20%. Does the winner change?
9. What is the single biggest technical risk in your design?
10. How does your PoC address that risk? Show me the data.

**Project Management and Feasibility**

11. What is your project budget, and how did you estimate it?
12. What is your longest lead-time item?
13. Show me your project schedule. Where is the critical path?
14. What happens to your schedule if the PoC fails?
15. Who is responsible for each major subsystem?

**Communication and Professionalism**

16. Is this report something you would send to a client? What would you improve?
17. Your figures lack axis labels (or units, or legends). Fix this.
18. I notice several team members have not spoken. [Name], can you explain the thermal analysis?
19. What question are you most afraid I will ask?
20. What is the one thing you wish you had done differently this semester?

## 4.3 CDR Readiness Assessment

Before approving a team for CDR, verify the following. If more than two items are incomplete, consider delaying the review.

- ☐ All PDR feedback items have been addressed (closed or explicitly deferred with justification).
- ☐ Detailed design is complete: CAD models, schematics, or analytical models are at fabrication-ready level.
- ☐ Bill of Materials is complete with part numbers, quantities, unit costs, and suppliers.
- ☐ Hazard Assessment is drafted and reviewed.
- ☐ Fabrication plan includes timeline, facility reservations, and responsible individuals.
- ☐ V&V test plan maps every requirement to a specific test procedure.
- ☐ Budget is updated with actual procurement costs.
- ☐ CDR report is a complete draft (not an outline or a collection of bullet points).

## 4.4 FDR Evaluation: What Distinguishes A-Level Work

**What separates an A-level FDR from a B-level FDR?**

**A-level FDR:**

- Every requirement in the VCRM is dispositioned with evidence (test data, analysis, or inspection records).
- The engineering calculation package demonstrates mastery: assumptions are stated, methods are justified, results include uncertainty analysis, and conclusions connect back to requirements.
- The team demonstrates engineering judgment—they made trade-offs, documented the reasoning, and can defend their decisions under questioning.
- Documentation quality is professional: consistent formatting, proper citations, clear figures with labeled axes, and coherent narrative flow.
- The team articulates lessons learned with specificity, not generic platitudes.

**B-level FDR:**

- Requirements are mostly verified, but some lack clear evidence or have weak test procedures.
- Calculations are correct but lack uncertainty analysis or stated assumptions.
- Design decisions are justified but only at a surface level.
- Documentation meets minimum standards but has inconsistencies or gaps.

## 4.5 Managing Q&A Sessions

**Drawing out quiet members:** Direct questions by name: "[Name], you led the structural analysis—can you walk us through the loading assumptions?" This is not punitive; it ensures individual accountability and gives every team member a chance to demonstrate competence.

**Redirecting off-topic answers:** When a student veers into tangential detail, use: "That is interesting, but let us come back to the specific question: does the test data confirm the requirement is met?" Be direct without being dismissive.

**Handling answers the team does not know:** It is acceptable—and even commendable—for a team to say "We do not have that data yet; we plan to test that in Sprint X." It is not acceptable for the answer to be "We had not thought about that." The former shows planning; the latter shows a gap.

## 4.6 Calibrating Expectations Across PAs

Consistency across PAs is essential for fairness. If one PA considers an FDR report acceptable and another considers the same quality level insufficient, students receive mixed signals and grade disputes increase.

Attend PA calibration sessions before each design review cycle. Bring a sample deliverable from one of your teams (anonymized if needed) and compare your assessment with other PAs. The CF facilitates these sessions to establish shared expectations.

# 5 Student Misconceptions by Topic

*Reference | Course Text: Chapters 3, 4, 5, 6, 7, 14, 18, 19, 22*

Students arrive in Capstone with strong technical skills but predictable gaps in engineering practice. This section catalogs the most common misconceptions by topic area, how to identify them, and how to address them. Recognizing these patterns early allows you to intervene before they become entrenched in deliverables.

## 5.1 Requirements

**Misconception:** Confusing specifications with requirements.
*How it appears:* "The system shall use an Arduino Mega 2560" instead of "The system shall process sensor inputs at a minimum rate of 100 Hz."
*How to address:* Ask: "Why an Arduino? What performance need drives that choice?" Redirect to the functional need. The hardware choice is a design decision, not a requirement.

**Misconception:** Requirements do not need to be verifiable.
*How it appears:* "The system shall be user-friendly" or "The device shall be reliable."
*How to address:* For every requirement, ask: "How will you prove this is met?" If they cannot describe a test, the requirement needs rewriting.

**Misconception:** The client's wish list is the requirements list.
*How it appears:* A verbatim copy of the client brief rephrased as "shall" statements with no engineering analysis or prioritization.
*How to address:* Have the team classify requirements as Must-Have, Should-Have, or Nice-to-Have. Ask which requirements they would cut if budget or schedule were halved.

## 5.2 Scope

**Misconception:** Saying yes to everything the client asks for.
*How it appears:* An SOW with no "Out of Scope" section, or one that lists only trivial exclusions.
*How to address:* Ask: "What will you *not* deliver?" A well-scoped project has clear boundaries. The "Out of Scope" section is as important as the "In Scope" section.

**Misconception:** Scope can be expanded mid-project without consequences.
*How it appears:* The team agrees to additional client requests without updating the schedule, budget, or requirements.
*How to address:* Introduce the concept of a scope change request. Every addition requires a corresponding trade-off: more time, more money, or something else removed.

## 5.3 Testing

**Misconception:** Testing only what works (testing theater).
*How it appears:* Test results show 100% pass rate, but the tests were designed to confirm the design rather than challenge it. No edge cases, no stress tests, no failure-mode testing.
*How to address:* Ask: "What did you test that you expected to fail?" and "What are the boundary conditions of your design?" Good testing tries to break things.

**Misconception:** Testing can wait until the build is "done."
*How it appears:* No V&V activity until Sprint 12. The team plans to "test everything at the end."
*How to address:* Point to the Course Text, Ch. 6. Testing should begin as soon as the first subsystem is functional. Every week of testing margin is one potential redesign cycle.

## 5.4 Budget

**Misconception:** Zero contingency in the budget.
*How it appears:* A budget that accounts for exactly the components needed with no margin for failed parts, design changes, or price fluctuations.
*How to address:* Industry practice is 10–20% contingency for well-defined projects. Ask: "What happens to your budget if a motor burns out during testing and you need to reorder?"

**Misconception:** Ignoring consumables and incidentals.
*How it appears:* BOM omits fasteners, wire, solder, adhesives, sandpaper, and 3D printing filament.
*How to address:* Have the team walk through the CAD assembly and fabrication steps. Every material they touch needs a cost line, even if it comes from shared lab stock.

## 5.5 Communication

**Misconception:** Dumping information instead of communicating.
*How it appears:* Reports that present every calculation performed without a narrative explaining why the calculation matters or what it means for the design. Presentations that are dense walls of text.
*How to address:* Ask: "What is the one thing you want the reader to take away from this section?" Every section of a report should answer a design question, not just display work.

**Misconception:** The PA or client will read between the lines.
*How it appears:* Critical decisions are undocumented because "everyone on the team knows."
*How to address:* Apply the "bus test": if one team member were unavailable, could someone else reconstruct the reasoning from the documentation alone?

## 5.6 Ethics

**Misconception:** Ethics is a checkbox exercise, not a design input.
*How it appears:* The ethics section of the report is a two-paragraph afterthought listing generic professional codes without connecting them to specific design decisions.
*How to address:* Ask: "Name one design decision on this project that has an ethical dimension." Push for specificity: whose safety, privacy, or welfare is affected by this design, and how did that influence your choices?

**Misconception:** If it is legal, it is ethical.
*How it appears:* Team dismisses ethical concerns by citing compliance with regulations.
*How to address:* Regulations define the floor, not the ceiling. Ask: "Who could be harmed by this design even if every regulation is followed?" Reference the FDVSE framework (Course Text, Ch. 5) for structured ethical analysis.

## 5.7 Sustainability

**Misconception:** Generic sustainability statements instead of project-specific analysis.
*How it appears:* "Our project is sustainable because we used recycled materials." No lifecycle analysis, no quantitative comparison of alternatives, no connection to the FDVSE framework.
*How to address:* Ask: "Compared to the alternative approach, what is the quantitative environmental impact difference?" Sustainability claims must be specific, measurable, and comparative—not aspirational platitudes. See Course Text, Ch. 5.

**Misconception:** Sustainability only means environmental impact.
*How it appears:* No consideration of economic sustainability (can the client maintain this after handoff?) or social sustainability (does this solution work for all intended users?).
*How to address:* Broaden the conversation using the FDVSE framework. Sustainability encompasses environmental, economic, and social dimensions.

# 6 Grading and Assessment

*Reference | Course Text: Chapters 11, 13*

Grading in Capstone Design is inherently more complex than in a lecture course. You are assessing engineering judgment, teamwork, and professional conduct—not just deliverable completeness. This section explains the grading structure, rubric philosophy, and common calibration challenges.

## 6.1 How Final Grades Are Determined

Final grades are determined by the CF using a combination of inputs:

1. **PA input.** Your assessment of team performance across sprints, design reviews, and deliverables. This is the most heavily weighted qualitative input.
2. **Deliverable scores.** Rubric-based scores on reports, presentations, and assignments (graded on Canvas).
3. **Feedback Loop.** Peer evaluations and Individual Performance Evaluations that differentiate individual contributions within a team.
4. **CF observation.** The CF's direct observation at design reviews and through course-level data.

The CF synthesizes these inputs into final grades. PAs do not assign final grades, but your input is the primary basis for the qualitative assessment. Take this responsibility seriously.

Do not promise or imply specific grades to students. You may describe quality levels ("this is strong work," "this needs significant improvement"), but grade assignment is the CF's responsibility. If a student presses you for a grade commitment, redirect them to the syllabus and the CF.

## 6.2 Rubric Philosophy

The grading rubrics evaluate two dimensions:

1. **Engineering quality.** Does the work demonstrate sound engineering judgment? Are tradeoffs analyzed with data? Are assumptions stated and justified? Are conclusions supported by evidence?
2. **Professional execution.** Is the deliverable complete, well-organized, properly formatted, and free of errors? Would you be comfortable sending this to a client?

When reviewing deliverables, ask yourself: "Would I accept this from a junior engineer on my team?" This framing calibrates expectations appropriately—better than a student report, but not yet at the level of a licensed professional engineer.

## 6.3 Assessing Engineering Calculation Packages

The engineering calculation package is where teams demonstrate analytical rigor. An A-level calculation package includes:

- **Stated assumptions.** Every assumption is listed, and the team can justify why each is reasonable.
- **Method justification.** The team explains why they chose this analytical method (FEA, hand calc, empirical correlation, etc.) and acknowledges its limitations.
- **Uncertainty analysis.** Results include uncertainty bounds or sensitivity analysis. A single point estimate without uncertainty is incomplete.
- **Validation.** Where possible, the analysis is compared against experimental data, published results, or an independent calculation method.
- **Design connection.** The conclusion explicitly connects the analysis result to a requirement or design decision. "The factor of safety is 2.3, which meets the requirement of $\geq$2.0" is useful. "The maximum stress is 145 MPa" without context is not.

## 6.4 Differentiating Individual Contributions

Team grades are the starting point, but individual grades may differ. Use the following evidence to differentiate:

- **Feedback Loop peer evaluations.** Cross-reference peer scores with your own observations. Consistent low scores from multiple peers are a reliable signal.
- **Sprint review participation.** Who answers questions with depth? Who defers to others on their own assigned topics?
- **Deliverable ownership.** Review version histories and commit logs. Who is writing, who is editing, and who is absent?
- **Meeting attendance and engagement.** Chronic absence or passive attendance is a differentiator.
- **PA Individual Performance Evaluation.** Your direct assessment of each student, completed at scheduled intervals throughout the semester.

Keep brief notes after each sprint review: one sentence per student summarizing their contribution that sprint. These notes are invaluable at the end of the semester when you need to differentiate grades and cannot rely on memory alone.

## 6.5 Handling Grade Disputes

Grade disputes occasionally arise, particularly when individual grades differ from team grades. Follow this process:

1. **Listen first.** Let the student explain their concern fully before responding.

2. **Refer to evidence.** Point to specific deliverable feedback, peer evaluation scores, and your observation notes. Avoid subjective characterizations without supporting data.
3. **Escalate to the CF.** If the student is not satisfied with your explanation, direct them to the CF. The CF makes final grade determinations and handles formal appeals.
4. **Do not change grades unilaterally.** Any grade changes go through the CF.

### 6.6 The Feedback Loop: Interpretation and Application

The Feedback Loop peer evaluation system generates quantitative scores (1–5 scale) and qualitative comments from teammates. Here is how to interpret and apply the data:

**Interpreting scores:**

- **4.0–5.0:** Strong contributor. Peer perception aligns with high performance.
- **3.0–3.9:** Adequate contributor. May have specific areas for improvement but is pulling their weight.
- **Below 3.0:** Consistent underperformance perceived by peers. Cross-reference with your own observations before acting.

**Applying scores:**

- A single low evaluation may reflect interpersonal friction, not performance. Look for *patterns* across multiple evaluation cycles.
- If peer scores and your observations diverge significantly, investigate. The team may be shielding a struggling member, or there may be a personality conflict distorting scores.
- Share aggregate trends (not individual scores) with the student: "Your peer evaluations suggest that your teammates would like to see more proactive communication."

A pattern of low Feedback Loop scores (below 3.0) combined with PA observation of non-participation can result in individual grade reduction, PIP initiation, or course failure—regardless of team deliverable scores. Document your observations throughout the semester.

## 7 Managing Team Issues

*Reference | Course Text: Chapter 13*

Team dysfunction is the most common source of PA intervention requests. Most issues are predictable and manageable if identified early. This section provides frameworks for diagnosis, escalation, and resolution.

### 7.1 The Tuckman Model Applied to Capstone

Teams progress through Forming, Storming, Norming, and Performing. In Capstone Design, the timeline is compressed:

- **Forming (Sprints 0–1):** Polite, cautious, low conflict. Teams avoid disagreement to maintain harmony. This is normal.

- **Storming (Sprints 3–5):** Conflict emerges as the workload increases and design decisions require trade-offs. Personality differences surface. This is the highest-risk phase.
- **Norming (Sprints 5–8):** Team establishes working norms. Roles solidify. Conflict is managed, not eliminated. PDR often serves as a unifying forcing function.
- **Performing (Sprints 9–14):** Effective teams reach this phase and execute efficiently. Struggling teams may never leave Storming.

Storming is healthy and expected. Your role during Storming is not to eliminate conflict but to ensure it stays productive. Productive conflict is about design decisions and priorities. Destructive conflict is personal, disrespectful, or passive-aggressive. Intervene in the latter; facilitate the former.

## 7.2 Identifying Underperformers Early

Watch for these indicators starting in Sprint 2:

- **Missed deadlines.** One slip is human. A pattern of two or more missed sprint deadlines signals disengagement.
- **Absent from meetings.** Missing team meetings or client meetings without advance notice.
- **No commits or contributions.** Version history and repository logs show minimal or zero contributions from the individual.
- **Cannot explain their own work.** When asked about their assigned subsystem at a sprint review, they defer to another team member.
- **Peers stop relying on them.** Other team members quietly redistribute the underperformer's work without telling you. This often surfaces in peer evaluations.

If a team member has contributed nothing visible in two consecutive sprints, do not wait for the next peer evaluation. Address it directly with the individual in a one-on-one conversation.

## 7.3 Escalation Path

Follow this escalation sequence. Each step should be documented.

1. **Team conversation.** Encourage the team to address the issue internally first. Healthy teams can resolve most issues through direct conversation. Ask the team: "Have you talked to [Name] about this directly?"
2. **PA-mediated conversation.** If the team conversation fails or the team is unwilling to have it, facilitate a structured conversation. Meet with the team (or the individuals involved) and establish specific, measurable expectations for the next sprint. Document the agreement.
3. **PA one-on-one.** For persistent issues, meet privately with the underperforming student. Be direct: "Your teammates and I have observed [specific behaviors]. Here is what needs to change by [specific date]." Document this conversation.
4. **CF intervention.** If the PA-mediated conversation does not produce change within one sprint, escalate to the CF. The CF may initiate a Performance Improvement Plan (PIP), reassign team roles, or take other corrective action.

Document every escalation step with dates, attendees, and specific commitments made. This documentation protects you, the student, and the program if the situation becomes a formal dispute.

## 7.4 The Bus Factor Problem

The "bus factor" is the number of team members who could be unavailable before the project stalls. A bus factor of one means all critical knowledge resides in a single person.

**Symptoms:**

- Only one person can explain the firmware, the CAD model, or the test procedure.
- One team member is CC'd on every client email; others are not.
- Team meetings cannot proceed when a specific member is absent.

**Remediation:**

- Require cross-training: each subsystem should have a primary and a secondary owner.
- Require documentation of all critical procedures (not just the final report—working documents).
- At sprint reviews, ask questions to the secondary owner, not the primary, to test knowledge transfer.

## 7.5 When to Involve the CLT Director

Escalate to the CLT Director (not just the CF) when:

- The issue involves the client relationship (client is unresponsive, client is making unreasonable demands, client contact has changed).
- The issue involves safety concerns in fabrication spaces (InnoHub, labs).
- The issue involves facilities, equipment access, or procurement logistics beyond normal channels.
- A team member raises a concern about harassment, discrimination, or hostile behavior (the CLT Director can involve institutional resources).

## 7.6 Personality Conflicts vs. Performance Issues

Distinguish between these two categories, as the intervention differs:

**Personality conflict vs. performance issue**

**Personality conflict:** All members are contributing, but interpersonal friction makes collaboration difficult. Two members disagree on approach and the disagreement has become personal. *Intervention:* Facilitate a structured conversation focused on shared goals. Establish communication norms. Remind the team that they do not need to be friends; they need to be professional.

**Performance issue:** One or more members are not meeting commitments, regardless of interpersonal dynamics. Deadlines are missed, work quality is below standard, or attendance is poor.

*Intervention:* Follow the escalation path above. Performance issues require specific, measurable expectations and consequences.

When a team reports a "personality conflict," investigate whether it is actually a performance issue in disguise. Often, interpersonal friction originates from one member not carrying their share of the workload, and the resulting resentment manifests as personal conflict.

# 8 Facilitating the FDVSE Value Assessment

*Reference | Course Text: Chapter 5*

The FDVSE framework—Feasibility, Desirability, Viability, Sustainability, and Ethical Responsibility—provides a structured approach to evaluating whether a design is not just technically sound but also worth pursuing. This section covers how to facilitate FDVSE discussions with your teams and assess FDVSE quality in deliverables.

## 8.1 Overview of the FDVSE Framework

FDVSE asks five questions about a proposed design:

1. **Feasibility.** Can we build it? Is it technically achievable with available resources, skills, and timeline?
2. **Desirability.** Do stakeholders want it? Does it solve a real problem that people care about?
3. **Viability.** Can it sustain itself? Is there a viable business model, maintenance plan, or operational pathway?
4. **Sustainability.** What are the environmental, economic, and social impacts across the lifecycle?
5. **Ethical Responsibility.** Who is affected, and are their interests adequately considered? What are the potential harms, and how are they mitigated?

FDVSE is not a sequential checklist—it is an integrative lens. Each dimension interacts with the others. A design that is feasible but not viable, or desirable but not ethical, has not been fully evaluated. Encourage teams to discuss the tensions *between* dimensions, not just within each one.

## 8.2 Integration with the PDR/CDR/FDR Gate Structure

FDVSE assessment should deepen as the project progresses through gates:

- **PDR:** Teams should demonstrate that they have *considered* all five dimensions. The assessment may be preliminary, but it should not be absent. Focus on Feasibility and Desirability—the team should have evidence that the concept is technically achievable and that stakeholders want it.
- **CDR:** Teams should present a more rigorous FDVSE analysis. Viability and Sustainability dimensions should now include quantitative data where possible (e.g., lifecycle cost estimates, energy consumption comparisons, maintenance requirements).
- **FDR:** The FDVSE assessment should be comprehensive and evidence-based. Ethical Responsibility analysis should reference specific design decisions and their consequences, not generic statements about professional codes.

## 8.3 Session Facilitation Notes

When facilitating an FDVSE discussion (whether in a sprint review, design review, or dedicated session), use the following structure:

1. **Set the frame (2 minutes).** Remind the team that FDVSE is not a grading exercise—it is a design tool. The goal is to surface risks and opportunities, not to produce polished prose.
2. **Walk through each dimension (3–5 minutes each).** For each dimension, ask the guiding questions below. Let the team discuss before you offer observations.
3. **Identify tensions (5 minutes).** Ask: "Where do two dimensions pull in opposite directions?" For example: the most feasible option may have the worst sustainability profile. The most desirable feature may raise ethical concerns. These tensions are where engineering judgment lives.
4. **Determine action items (3 minutes).** For each dimension rated Yellow or Red, identify one concrete action the team will take before the next gate.

## 8.4 Guiding Questions by Dimension

**Feasibility**

- What is the single biggest technical risk, and how will you retire it?
- Do you have the skills on the team to execute this design, or do you need to learn something new?
- Is this achievable within the course timeline and budget?
- What is your fallback if the primary approach fails?

**Desirability**

- Who are the end users, and how did you validate that they want this?
- What evidence (beyond the client's statement) supports the need for this solution?
- If the end user is different from the client, whose needs take priority when they conflict?
- How will you measure user satisfaction with the final product?

**Viability**

- After handoff, who will maintain and operate this system? Do they have the necessary skills and resources?
- What are the ongoing costs (maintenance, consumables, energy, labor)?
- What is the expected operational lifetime?
- If the client cannot maintain this, what is the alternative?

**Sustainability**

- What is the environmental impact of this design compared to the alternative or the status quo?
- Have you considered the full lifecycle: materials, manufacturing, use, and end-of-life?
- Is this solution economically sustainable for the client organization over the intended lifetime?
- Does this solution create or exacerbate any social inequities?

**Ethical Responsibility**

- Who could be harmed by this design, directly or indirectly?
- What data does the system collect, and how is it protected?
- Are there dual-use concerns (could this be misused in ways you did not intend)?
- If this design fails, what are the consequences, and for whom?
- Name one design decision on this project that has an ethical dimension. How did ethics influence the decision?

## 8.5 Common Student Struggles with Ethical Responsibility

The Ethical Responsibility dimension is consistently the weakest in student FDVSE assessments. Common struggles include:

- **Defaulting to codes of ethics.** Students cite the NSPE or ASME code of ethics without connecting specific code provisions to specific design decisions. Push for concrete application.
- **Confusing legality with ethicality.** Students assume that if their design meets regulations, it is ethical. Regulations are the floor, not the ceiling.
- **Inability to identify affected parties.** Students consider only the direct user and the client, ignoring bystanders, maintainers, communities near manufacturing or disposal sites, and future generations.
- **Treating ethics as an afterthought.** The ethics analysis appears in the last paragraph of the report, disconnected from design decisions. Effective ethical analysis should be woven into the design narrative.

When students struggle with ethical analysis, try the "newspaper test": "If the local newspaper ran a story about your design going wrong, what would the headline be? Who would be quoted? What would they say?" This concrete scenario helps students move from abstract principles to specific consequences.

## 8.6 Assessing FDVSE Quality in PDR Presentations

At PDR, evaluate FDVSE using the following criteria:

| Level | Description | Indicator |
|---|---|---|
| Strong | All five dimensions addressed with project-specific evidence. Tensions between dimensions identified and discussed. | Team can articulate trade-offs between dimensions without prompting. |
| Adequate | All five dimensions mentioned. Some project-specific content, but analysis is surface-level or relies on generic statements. | Team addresses dimensions when asked but does not proactively discuss tensions. |
| Weak | One or more dimensions missing or treated as a checkbox. No connection between FDVSE analysis and design decisions. | Team cannot explain how FDVSE influenced any design choice. |

# 9 Project Health Monitoring

*Reference | Course Text: Appendix B*

Detecting a struggling project early is the most valuable skill a PA can develop. By the time a project is visibly failing (missed design review, client complaint, team member dropout), the window for effective intervention has often closed. This section provides tools for ongoing health monitoring.

## 9.1 The Project Health Check as a Management Tool

The Course Text (Appendix B) includes a Project Health Check: a self-assessment tool that asks teams to rate themselves Green/Yellow/Red across twelve common failure patterns. While designed for student self-reflection, the Health Check is equally valuable as a PA management tool.

**Recommendation:** Make the Project Health Check a mandatory, graded deliverable at the end of each semester (Sprint 6 and Sprint 14). Require teams to complete it honestly and discuss the results with you. The act of self-diagnosis is as valuable as the resulting data.

## 9.2 Green / Yellow / Red Indicators

Use the following general framework to assess project health at any point in the semester:

**Green: On Track**

- Deliverables are submitted on time and at or above the expected quality level.
- Team meets regularly and all members participate actively.
- Client communication is proactive and documented.
- Schedule shows margin for testing and iteration.
- The team can articulate their current risks and mitigation strategies.

**Yellow: Caution**

- One or two deliverables have been late or below expected quality.
- Team meetings are inconsistent, or one member's attendance is spotty.
- Client communication is reactive (responding to client inquiries rather than initiating).
- Schedule shows no margin; the critical path has no float.
- The team acknowledges risks but has no concrete mitigation plan.

**Red: Intervention Required**

- Multiple deliverables are late or substantially below quality expectations.
- Team conflict is unresolved and affecting productivity.
- Client has expressed concern or is unresponsive due to loss of confidence.
- Schedule shows the team cannot complete the project as scoped.
- The team cannot articulate what they are doing next or why.

## 9.3 When to Intervene vs. When to Let the Team Struggle

Not all struggle is bad. The distinction between productive and destructive struggle is central to effective PA mentoring.

**Productive struggle** (let it continue):

- The team is wrestling with a genuine technical challenge and making incremental progress.
- Team members are debating design trade-offs with data and reasoning (even if the debate is heated).
- The team has identified a problem and is working through a solution, even if slowly.
- Failure of a PoC or test provides learning that redirects the design.

**Destructive struggle** (intervene):

- The team is stuck on the same problem for more than one sprint with no visible progress.
- Conflict has become personal, and team members are avoiding each other rather than collaborating.
- The team does not recognize they are in trouble (no self-awareness of project health).
- Safety is at risk: fabrication without a Hazard Assessment, skipping test procedures, or ignoring equipment limitations.
- The team has given up on a deliverable or a requirement without discussing it with you.

When you decide to let a team struggle productively, tell them: "I can see this is challenging, and I think working through it will teach you more than if I give you the answer. I am here if you get completely stuck." This acknowledges the difficulty without removing agency. Check in at the next meeting to verify progress.

## 9.4 Documentation Debt: The Leading Indicator

If you track only one health metric, track documentation debt.

Documentation debt is the number one indicator of a project in trouble. Teams that build without writing—deferring all report sections, test logs, and design rationale to "later"—produce rushed, low-quality final deliverables and often miss critical details because they are reconstructing decisions from memory rather than from contemporaneous records.

**How to monitor:**

- At each sprint review, ask: "What report sections have you written this sprint?"
- Check the team's shared document (Google Docs, Overleaf, or similar) for recent activity.
- Compare the number of pages drafted against the number of pages expected at this point in the semester.
- If the team is more than two sprints behind on documentation, flag it as Yellow and establish a remediation plan.

Establish a documentation expectation in Sprint 0: report sections should be drafted during the sprint in which the work is performed, not retroactively at the end of the semester. Reinforce this expectation at every sprint review.

# 10 Track-Specific PA Guidance

*Reference | Course Text: Chapters 7, 10, 12*

Not all capstone projects follow the same trajectory. The course accommodates four project tracks—Build, Paper, Competition, and Hybrid—each with distinct PA considerations. This section highlights what to watch for in each track.

## 10.1 Build Track

Build Track projects produce a physical prototype or system. This is the most common track and the one with the most logistical complexity.

**Key PA considerations:**

- **Fabrication timeline awareness.** Physical builds are constrained by material lead times, InnoHub availability, and fabrication sequencing. A schedule that shows "fabrication" as a single three-week block is not a real schedule. Push for a detailed fabrication plan with subtasks and dependencies.
- **InnoHub coordination.** Teams must reserve InnoHub time and equipment in advance. If multiple teams need the same machine (e.g., laser cutter, CNC mill) during the same sprint, conflicts arise. Encourage early reservations.
- **Procurement pacing.** Purchasing at Mines follows institutional procedures (see Course Text, Ch. 15). Lead times for purchase orders, vendor quotes, and shipping can exceed two weeks for routine items and six or more weeks for specialty components. Teams that do not initiate procurement by Sprint 6 will face delays in SDII.

A Build Track team that enters SDII without having ordered long-lead components (PCBs, custom machined parts, specialty sensors) is already behind schedule. Verify procurement status at the Sprint 6 review.

- What is your longest lead-time component, and when does it need to arrive?
- Have you reserved InnoHub time for your fabrication milestones?
- What is your integration sequence, and what interfaces have you tested?

## 10.2 Paper Track

Paper Track projects produce an analytical deliverable (feasibility study, design analysis, simulation, optimization) rather than a physical prototype. The PA challenge is ensuring that analytical rigor matches what Build Track teams demonstrate through physical testing.

**Key PA considerations:**

- **Analytical rigor equals build rigor.** Paper Track teams sometimes treat the project as a literature review with some calculations appended. Push for the same depth of engineering analysis that a Build Track team would invest in design, fabrication, and testing.
- **What a Paper Track PoC looks like.** The PoC for a Paper Track project is a preliminary analysis or simulation that retires the key technical uncertainty. For example: a mesh convergence study for an FEA model, a sensitivity analysis for an optimization, or a validation of a simulation against published experimental data.
- **V&V for analysis.** Verification and validation still apply. Verification: does the model solve the equations correctly (mesh convergence, time-step independence, unit consistency)? Validation: do the model results agree with reality (comparison to experimental data, published benchmarks, or independent methods)?

For Paper Track teams, replace the question "Show me your prototype" with "Show me your model validation." The intellectual rigor standard is the same; the artifact is different.

- How have you validated your model against experimental data or published results?
- What assumptions does your analysis rely on, and how sensitive are your results to those assumptions?
- If a decision-maker used your results to make a $100K investment, would you be confident in the recommendation?

### 10.2.1 Paper Track Sprint-by-Sprint PA Focus

Use the following as a checklist for what to probe at each major sprint milestone when advising a Paper Track team.

| Sprint | What the PA Should Be Looking For |
|---|---|
| Sprint 2–3 | Are requirements *quantitative*? Do they include convergence criteria, accuracy targets, and validation benchmarks? Requirements that read "develop a model" or "analyze the system" are too vague. Push for measurable pass/fail criteria: mesh convergence within X%, agreement with published data within Y%, sensitivity sweep across parameter range Z. |
| Sprint 4 (PoC) | Is there actual computational evidence? A converging mesh study, a validated sub-model, or a parametric sweep that confirms viability? "I set up the software" or "the simulation runs without crashing" is not a PoC. The result must be *quantitative* and must retire a key technical uncertainty. |
| Sprint 5 (PDR) | Is the model architecture clearly described? Are preliminary results shown with appropriate uncertainty? Is the validation plan specified—what data will the model be compared against, and what constitutes acceptable agreement? Are all major assumptions documented and justified? |
| Sprint 8 (CDR) | Is the full model producing results? Has a sensitivity analysis been performed to identify which parameters drive the output? Are results validated against published data, experimental measurements, or independent analytical methods? Has an independent verification check been conducted (e.g., a hand calculation on a limiting case)? |
| Sprint 12–13 | Is the V&V documentation complete? Is there an assumptions log with justification for every major assumption? Is a user guide or methodology description included so that another engineer could reproduce the work? Are all results reproducible from documented inputs? |
| Sprint 14 (FDR) | Is the technical package complete? Are recommendations defensible and tied to quantitative results? Are model limitations explicitly acknowledged, including the parameter ranges and conditions under which the results should *not* be applied? |

### 10.2.2 Assessing Paper Track Engineering Judgment

Paper Track teams must demonstrate engineering judgment equivalent to what Build Track teams show through fabrication decisions, testing, and iterative redesign. The form is different, but the standard is the same. Use these four lenses when evaluating a Paper Track team's maturity.

- **Assumption quality.** Are the team's assumptions physically reasonable and well-justified? Every simplification should be accompanied by a rationale: why is this assumption acceptable, and what is the expected impact on results? Arbitrary assumptions ("we assumed a friction coefficient of 0.3 because it seemed reasonable") indicate weak engineering judgment.
- **Sensitivity awareness.** Does the team know which parameters drive their results? A team with strong judgment can tell you: "Our output is most sensitive to X and Y, relatively insensitive to Z, and we have bounded the uncertainty in X by doing a parametric sweep." A team that cannot articulate parameter sensitivity does not yet understand their own model.
- **Validation rigor.** Is the team comparing their results to appropriate benchmarks—published experimental data, analytical solutions for limiting cases, or independent computational results? "The model runs and produces numbers" is not validation. The team must demonstrate *quantitative* agreement (or explain discrepancies) against a credible reference.
- **Professional applicability.** Could a practicing engineer use this deliverable to make a real decision? If the answer is no—because the assumptions are unclear, the validation is missing, or the limitations are unstated—then the work does not meet the capstone standard, regardless of how sophisticated the model appears.

For Paper Track teams, the CDR readiness question is not "Is the prototype fabrication-ready?" but "Are the model results trustworthy enough to inform an engineering decision?" If the team cannot articulate the bounds of their model's validity, they are not ready for CDR.

A Paper Track team that describes their PoC as "setting up the software" or "getting the simulation to run" has not demonstrated a PoC. The PoC must show a quantitative result that retires a key technical uncertainty—a converging mesh study, a validated sub-model, or a parametric sweep that confirms the approach is viable.

## 10.3 Competition Track

Competition Track projects are driven by external competition rules, deadlines, and judging criteria. These projects introduce unique scheduling and scoping challenges.

**Key PA considerations:**

- **Mapping competition rules to course gates.** Competition deadlines rarely align with the course sprint structure. Help the team create a dual timeline showing both competition milestones and course gates (PDR, CDR, FDR). Identify conflicts early and plan accordingly.
- **Travel logistics.** Competitions often require travel. Travel planning at Mines involves institutional approval, funding requests, and logistical coordination (see Course Text, Ch. 16). Start travel planning at least 8 weeks before the event. Escalate funding or approval issues to the CLT Director.
- **External deadlines take precedence operationally.** If the competition submission deadline falls before CDR, the team must meet the competition deadline even if CDR preparation suffers. Work with the CF to adjust course gate timing if needed.
- **Scope management.** Competition rules often define requirements. The PA's role shifts from helping define scope to helping the team make strategic decisions about which competition criteria to prioritize given course constraints.

A Competition Track team that has not started travel planning by Sprint 4 (for a spring competition) will face a last-minute logistics scramble. Verify travel status early.

- Show me your dual timeline: competition milestones and course gates. Where are the conflicts?
- Which competition criteria will you prioritize, and which will you deprioritize?
- What is the status of travel approvals and funding?

### 10.3.1 Competition Track Timeline Management

The central challenge for Competition Track PAs is managing two timelines simultaneously: the course sprint structure and the external competition calendar. Neither timeline will accommodate the other, so the PA must help the team build an integrated plan that satisfies both.

**Creating the integrated timeline overlay.** In Sprint 0, have the team list every competition milestone (registration deadline, design report submission, tech inspection, competition dates) alongside every course gate (PDR, CDR, FDR). Plot both on a single Gantt chart or timeline. Identify every point of conflict—places where a competition deliverable is due within one week of a course gate, or where competition travel overlaps with a sprint review. Bring these conflicts to the CF in Sprint 1 so that gate timing can be adjusted before it becomes a crisis.

**When competition deadlines conflict with course gates.** This will happen. The general principle is: external deadlines take precedence operationally (you cannot ask a competition organizer to reschedule), but course gates take precedence for grading. Work with the CF to shift course gate timing by one to two weeks if needed. If a shift is not possible, help the team plan to complete the course deliverable early so that competition preparation does not crowd it out.

**Working with the CF on gate adjustments.** PAs do not have authority to move course gates unilaterally. Present the conflict to the CF with a proposed alternative date and a justification. Most CFs will accommodate reasonable requests if they are made early. Last-minute requests (less than two weeks before a gate) are much harder to approve.

**Sprint-by-sprint Competition Track PA focus:**

| Sprint | What the PA Should Be Looking For |
|---|---|
| Sprint 0–1 | Has the team read the *entire* competition rulebook, including appendices and safety requirements? Is there a compliance matrix mapping every rule to a design decision or verification method? Teams that skim the rules inevitably miss a critical requirement. |
| Sprint 2–3 | Do the requirements capture *both* competition rules and engineering requirements? Competition safety rules must be classified as MUST HAVE. Verify that the team is not treating competition rules as the only source of requirements—course-level engineering requirements (e.g., maintainability, documentation, testability) still apply. |
| Sprint 4 (PoC) | Does the PoC address the highest-risk competition requirement? For example, if the competition requires a specific performance threshold, the PoC should demonstrate feasibility of meeting that threshold, not just general system functionality. |
| Sprint 5 (PDR) | Is the competition compliance checklist part of the PDR presentation? Is competition registration confirmed? If registration requires a design report or abstract, is that deliverable on the team's timeline? |
| Sprint 6 | Has the team initiated travel planning? Mines institutional travel approvals require a minimum of 8 weeks. If travel planning has not started by Sprint 6, escalate to the CF immediately. Verify that funding sources have been identified and requested. |
| Sprint 8 (CDR) | Is the travel plan finalized (approvals submitted, flights/hotels identified, funding confirmed)? Is competition compliance nearly complete—does the team have a clear list of remaining compliance items with a plan to close each one? |
| Sprint 9–11 | Is the team testing against *competition-specific* criteria, not just general performance? Are they practicing for tech inspection (if applicable)? Is someone tracking rule updates and organizer communications? |
| Sprint 12–13 | Are competition logistics finalized (packing lists, shipping plans for equipment, travel itineraries)? Have practice runs or mock competitions been completed? Is the course documentation (final report, poster) also on track, or has competition prep crowded it out? |
| Sprint 14 (FDR) | Are competition results presented and analyzed regardless of placement? The FDR should include lessons learned, performance against competition criteria, and honest assessment of what worked and what did not. |

### 10.3.2 Managing Competition Disappointment

Competition outcomes are inherently uncertain. A well-engineered design can lose due to judging subjectivity, rule interpretations, equipment failures during transport, or simply facing stronger competitors. PAs must be prepared to help teams navigate the emotional and academic aftermath of competition results.

- **Teams that perform poorly at competition.** The most important PA intervention is helping the team refocus on course deliverables and learning outcomes. A poor competition result does not mean the project failed. Guide the team to analyze *why* things went wrong using engineering analysis (root cause, failure mode identification) rather than emotional narratives. The FDR should present this analysis honestly—it often produces more learning than a first-place finish.
- **Teams that neglect documentation for competition focus.** Some teams become so absorbed in competition preparation that they neglect course documentation (final report, poster, FDR preparation). Watch for this pattern starting around Sprint 9. If the team's documentation is falling behind while competition prep accelerates, intervene early and help them allocate effort to both streams.

- **Competition placement is not part of the grade.** Remind teams of this explicitly and repeatedly. The course evaluates engineering process, documentation quality, technical rigor, and professional development. A team that places last at competition but produces excellent documentation, demonstrates strong engineering reasoning, and learns from the experience can earn a top grade. Conversely, a team that wins the competition but submits a weak report has not met the course standard.
- **Extracting engineering learning from results.** Regardless of placement, help the team turn competition results into engineering data. What did the competition reveal about their design that testing did not? How did their performance compare to their predictions? What would they change with another design cycle? These questions produce the kind of reflection that distinguishes capstone from a class project.

If a Competition Track team has not confirmed registration and initiated travel planning by Sprint 6, escalate to the CF immediately. Mines travel approvals require 8 weeks minimum, and late travel requests create administrative burdens for the department.

Encourage Competition Track teams to attend all competition organizer webinars and info sessions. Rule interpretations often come from Q&A sessions, not from the written rulebook. Assign one team member as the "Competition Manager" responsible for tracking all competition communications.

## 10.4 Hybrid Projects

Hybrid projects combine elements of multiple tracks—for example, a Build Track project with a significant analytical component, or a Competition Track project that also produces a standalone deliverable for a client. These are the most complex to manage.

**Key PA considerations:**

- **Managing dual deliverable streams.** Hybrid projects require the team to produce deliverables for two audiences or two standards simultaneously. Help the team identify which deliverables serve both purposes (reducing duplication) and which are unique to each stream.
- **Preventing scope creep.** Hybrid projects are especially vulnerable to scope creep because two sets of requirements (course + competition, or course + client extension) create pressure to do everything. Enforce a clear scope boundary.
- **Balancing effort.** Watch for teams that invest disproportionately in one stream (e.g., the competition build) while neglecting the other (e.g., the course report). Both streams must meet quality standards.

For Hybrid projects, establish a clear priority framework in Sprint 0: which deliverable stream takes precedence when resources are constrained? Document this agreement with the team and the client. Revisit it at CDR.

#### 10.4.1 Common Hybrid Configurations

Most Hybrid projects fall into one of the following patterns. Recognizing the configuration early helps the PA anticipate the specific tensions and integration challenges that will arise.

- **Build + Paper.** A physical prototype with significant analytical validation—for example, a structural design that includes FEA validation, or a thermal system with a companion computational model. The PA challenge is ensuring that both the physical and analytical work streams meet their respective rigor standards and that the two inform each other (model predictions compared to test results, test results used to refine the model).
- **Competition + Client.** A competition entry that also serves a client need—for example, a SAE competition vehicle sponsored by an industry partner with specific performance requirements beyond the competition rules. The PA must help the team manage two sets of stakeholder expectations and ensure that client deliverables (reports, presentations, data packages) are not sacrificed for competition preparation.
- **Multi-disciplinary integration.** Projects requiring ME + EE + CS integration (or similar cross-disciplinary scope) where different team members produce fundamentally different deliverable types—mechanical drawings, circuit schematics, software repositories. The PA challenge is ensuring coherent system-level integration and documentation that captures the interfaces between disciplines, not just the individual subsystems.
- **Research + Build.** A faculty research project with a physical demonstrator component—for example, a novel mechanism concept that requires both theoretical analysis (kinematics, dynamics) and a proof-of-concept prototype. The PA must help the team balance the open-ended nature of research (where negative results are valuable) with the course requirement for concrete, demonstrable deliverables.

# 11 Purchasing and Travel Oversight

*Reference | Course Text: Chapters 15, 16*

As the PA, you are the approval authority for all team purchases and the facilitator for all travel planning. These responsibilities carry real financial consequences—unauthorized purchases create personal liability for students, and poorly planned travel wastes program funds. This section covers both domains.

## 11.1 Purchasing: Your Role as Approval Authority

Every purchase requires PA approval before the team spends money. This is not a rubber stamp—your approval confirms that the purchase is necessary, budgeted, and routed through the correct process. The CFO initiates the purchase; you authorize it.

#### 11.1.1 The Approval Workflow

1. The team identifies a need and selects a component or material.
2. The CFO presents the purchase to you: what, why, how much, which vendor, and where it fits in the budget.
3. You verify the purchase is justified, within budget, and from an approved vendor.

4. The team submits the appropriate form to Capstone Purchasing (CP): Purchase Form, Shopping List, or Reimbursement Form.
5. CP processes the order and the team picks up from Labriola 204.

Establish a simple approval channel early—email or Slack message with the item, vendor, cost, and budget line. A quick "Approved" reply creates a paper trail. Do not let teams rely on verbal approvals alone; undocumented approvals lead to disputes.

#### 11.1.2 Reviewing the Budget Tracker

The Budget Tracker is uploaded to Canvas monthly. Review it at least once per month and ask:

- Does actual spending match the approved budget categories?
- Are there unexpected line items or categories trending over budget?
- Has the team accounted for upcoming large purchases (custom PCBs, raw materials, shipping)?
- Is the remaining budget sufficient to complete the project?

Check the Budget Tracker at every sprint review during Sprints 8–11 (the build phase). This is when spending accelerates and overruns develop. A 30-second budget check during your meeting prevents surprises at FDR.

#### 11.1.3 Understanding the CP Workflow

Capstone Purchasing (CP) processes all orders. Three forms exist on Canvas:

- **Purchase Form**: standard orders from approved vendors.
- **Shopping List**: multiple items from the same vendor.
- **Reimbursement Form**: student-made purchases (limited; requires your pre-approval, an itemized receipt, and no sales tax).

Teams must email CP (capstonepurchasing@mines.edu) with: team name/number, submission date, vendor name, and form used. CP processing takes 3–5 business days. Orders from unapproved vendors require a separate vendor review that may add 1–2 weeks.

Remind your teams that CP staff process hundreds of orders per semester. Complete, accurate forms get processed faster. If a team complains about slow processing, check whether they submitted the form correctly first.

#### 11.1.4 Manufacturing Review Meeting Requirement

Build Track teams must complete the **Manufacturing Review Meeting** before CP will process orders for build materials. This is a required checkoff on the purchasing form. If a team submits a materials order without this checkoff, CP will reject it.

- Has the Manufacturing Review Meeting been completed?
- Does the BOM include all materials—including fasteners, consumables, and wiring?
- Are long-lead items ordered or submitted to CP?

### 11.1.5 Common Purchasing Pitfalls

Watch for these recurring problems across your teams:

**Unauthorized purchases.** A student buys components with personal funds without PA approval, then submits a reimbursement request. This violates the purchasing workflow and may result in non-reimbursement. Reinforce at Sprint 7: no purchase without prior approval, no exceptions.

**Missing tax exemption.** Mines is tax-exempt. Sales tax paid on purchases is **not reimbursable**. If a student buys locally without presenting the tax exemption certificate, the tax comes out of the team budget—or the student's pocket. Ensure the CFO has the certificate saved and accessible.

**Late procurement.** Teams that delay ordering long-lead components (custom PCBs, specialty sensors, raw materials) past Sprint 8 risk not receiving parts before Spring Break. Items ordered after break have almost no schedule margin. Push teams to order critical components early and check procurement status at every sprint review.

If a team's spending is trending significantly over budget or you discover unauthorized purchases, escalate to the CF. Budget overruns and purchasing violations are course policy issues that may require intervention beyond your PA authority.

### 11.1.6 When to Escalate Budget Concerns

Escalate to the CF when:

- A team's actual spending exceeds the approved budget by more than 15%.
- A student has made unauthorized purchases and is seeking reimbursement.
- A team needs a budget increase that exceeds your comfort level.
- A purchasing dispute arises between the team and CP that you cannot resolve.
- A team is not submitting Budget Tracker updates on schedule.

## 11.2 Travel: Facilitating Student Travel Planning

Competition teams, field survey projects, and client site visits may require travel. As the PA, you facilitate the planning, approve the travel request, and serve as the safety backstop. The students

handle logistics; you ensure the logistics are sound.

### 11.2.1 The Go/No-Go Decision

Before any travel spending begins, you must approve a **Go/No-Go decision** in coordination with the Capstone Leadership Team (CLT). The Go/No-Go considers:

- Is the trip essential to the project or competition?
- Is the travel budget available and realistic?
- Is the team prepared (prototype ready, competition paperwork complete)?
- How many travelers are needed? Determine the **travel roster**—the minimum number of people required for the mission.

Initiate the Go/No-Go conversation with CLT at least 8 weeks before the planned travel date. Late Go/No-Go decisions lead to expensive last-minute bookings and missed registration deadlines.

### 11.2.2 Approving Travel Requests

Once Go/No-Go is approved, the team's designated Travel Lead prepares a detailed budget. Before approving the travel budget, verify:

- Multiple lodging options with cost comparisons and links.
- Airfare quotes per traveler (through FROSCH if required by Mines policy).
- Ground transport plan: rental car, mileage for personal vehicles, or shared transportation.
- Per diem estimates for all travelers for all travel days.
- Registration and competition fees.
- Equipment shipping costs (outbound and return).

Require the Travel Lead to present the budget using the Budget Template on Canvas, with actual quotes—not guesses. Teams that present vague cost estimates invariably exceed their budget. If the budget is not detailed enough, send it back before approving.

### 11.2.3 Understanding Mines Travel Policies

Key policies you must enforce:

- **Carpooling required**: multiple personal vehicles for the same trip will not be reimbursed.
- **Mileage**: reimbursed at the Mines rate, calculated from 1500 Illinois St., Golden CO 80401. The team must save the Google Maps route as a PDF.
- **Fuel**: not reimbursable for personal vehicles (mileage rate covers fuel).
- **Meals**: itemized receipts required. **No alcohol on any receipt.** A receipt containing alcohol is rejected *in full*—no partial reimbursement.
- **Post-trip receipts**: must be submitted to CP within one month of trip completion.
- **Rental cars**: driver must be $\geq$21 years old with a valid license. Book through CP using the OneCard.

The alcohol receipt rule catches teams every semester. If any item on a receipt is an alcoholic beverage, the **entire receipt** is rejected. Remind teams before every trip: pay for alcohol separately on a personal card, or do not order alcohol on team travel.

#### 11.2.4 Competition Team Travel Logistics

Competition travel involves the largest budgets and most complex logistics in capstone. Key areas to oversee:

**Vehicle rental and ground transport.** Rental cars must be booked through CP using the OneCard. Verify the designated driver meets age and license requirements. For driving trips, ensure the team plans rest stops every 2–3 hours, limits driving to 10 hours per day, and switches drivers regularly.

**Hotel booking.** Two occupants per room to control costs. Book with a free cancellation policy. Verify proximity to the competition venue. Request late checkout on the last day if the event runs late.

**Registration and deadlines.** Competition registration often closes months before the event. Once Go/No-Go is approved, the Travel Lead must submit registration immediately. Track deadlines for tech inspection paperwork, safety forms, driver certifications, and other pre-event requirements.

**Equipment shipping.** If shipping the prototype or competition vehicle, the team must plan logistics well in advance: measure and weigh the shipment, get multiple carrier quotes, and ship at least 5 business days before the event. A prototype that arrives after the competition is a preventable disaster.

- Who is the designated Travel Lead, and do they have a detailed budget?
- Have all travelers verified their IDs match their booking names?
- What is the equipment shipping plan and timeline?
- What is the contingency if a flight is cancelled or equipment is delayed?

#### 11.2.5 Safety Considerations for Off-Campus Activities

You are responsible for safety oversight of all off-campus activities, including travel and client site visits.

- **Itinerary sharing**: require the team to share the complete travel itinerary with you before departure. Every traveler should have your phone number.
- **Emergency contacts**: for competition travel, identify the nearest hospital to the venue. Ensure every traveler has the PA's phone number, the campus emergency number, and the hotel address.
- **Personal vehicle safety**: drivers must follow the Mines driving policy (available from EHS). Adequate sleep before long drives. Valid, current insurance.
- **Client site visits**: before visiting a client site or field location, the team must assess site-specific hazards (construction, confined space, chemical, biological). Ask the client about required PPE and safety orientation before the visit.

- **Weather and environment**: for outdoor or remote field work, verify the team has checked weather conditions, has cell coverage or a satellite communicator, carries a first aid kit, and has informed someone of their location and expected return time.

A team plans an off-campus activity (site visit, field test, competition) without informing you. Any off-campus activity related to the project requires your awareness and, for travel, your explicit approval. If you discover a team visited a client site or conducted a field test without telling you, address it immediately.

#### 11.2.6 Required Documentation and Pre-Approval Process

The travel planning timeline requires action well before the trip:

- **8–12 weeks before**: Go/No-Go conversation with CLT. Identify the travel need and designate the Travel Lead.
- **6–8 weeks before**: Travel Lead develops the budget. Research lodging, airfare, and ground transport options. Register for the event.
- **4–6 weeks before**: submit the travel budget to you and CP for approval. Book lodging and begin airfare booking. Confirm the travel roster.
- **2–4 weeks before**: finalize all bookings. Prepare the equipment packing list. Verify all traveler documents (IDs, registration, competition paperwork).
- **1 week before**: pre-trip meeting with the team. Distribute the itinerary. Verify all reservations. Confirm emergency contacts.
- **Within 1 month after**: all receipts and reimbursement forms submitted to CP. Budget Tracker updated with actual travel costs.

For international travel, additional requirements apply: valid passports (6+ months remaining), registration with the Mines Office of Global Education (4+ weeks before departure), and verification of export control restrictions on any technical equipment.

One week before departure, ask the Travel Lead to walk you through the pre-trip checklist: confirmed reservations (names match IDs), rental car and driver verification, competition registration and pre-event paperwork, equipment shipping status, emergency contact distribution, and printed copies of all itineraries and confirmations.

*This guide is a companion to the EDNS 491/492 Course Text and Student Guide. For detailed methodology, technical reference, and worked examples, consult the Course Text chapters referenced in each section above.*